```
abstract class EnergySource {
    String sourceId;
    String sourceName;
    double energyGenerated;

    EnergySource(String sourceId, String sourceName, double
energyGenerated) {
        this.sourceId = sourceId;
        this.sourceName = sourceName;
        this.energyGenerated = energyGenerated;
    }


    abstract double calculateEfficiency();


    void displayDetails() {
        System.out.println("Source ID: " + sourceId);
        System.out.println("Source Name: " + sourceName);
        System.out.println("Energy Generated: " + energyGenerated + "
kWh");
        System.out.printf("Efficiency: %.2f\n", calculateEfficiency());
        System.out.println("-----------------------------------");
    }
}


class SolarEnergy extends EnergySource {

    SolarEnergy(String sourceId, String sourceName, double
energyGenerated) {
        super(sourceId, sourceName, energyGenerated);
    }


    double calculateEfficiency() {
        return (energyGenerated / 5000.0) * 100.0;
    }
}


class WindEnergy extends EnergySource {

    WindEnergy(String sourceId, String sourceName, double energyGenerated)
{
        super(sourceId, sourceName, energyGenerated);
    }
```

```
    double calculateEfficiency() {
        return (energyGenerated / 8000.0) * 100.0;
    }
}


public class EnergyManagementSystem {
    public static void main(String[] args) {

        EnergySource ref;

        ref = new SolarEnergy("SOL-101", "Sahara Solar Park", 3750.0);
        ref.displayDetails();


        ref = new WindEnergy("WND-202", "Alta Wind Energy Center",
5600.0);
        ref.displayDetails();
    }
}
```

OUTPUT:
Source ID: SOL-101
Source Name: Sahara Solar Park
Energy Generated: 3750.0 kWh
Efficiency: 75.00

Source ID: WND-202
Source Name: Alta Wind Energy Center
Energy Generated: 5600.0 kWh
Efficiency: 70.00
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